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# TASK 1: DATA CLEANING & STANDARD SCALING FOR DISTANCE-BASED ALGORITHMS
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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

# 1. Load the Palmer Penguins dataset from your verified file 'ReadMe.csv'
df = pd.read_csv('ReadMe.csv')

# 2. Isolate continuous numeric columns requested for morphology analysis
numeric_features = ['bill_length_mm', 'bill_depth_mm', 'flipper_length_mm', 'body_mass_g']

# 3. Clean dataset by removing rows with missing values in these key features
df_clean = df.dropna(subset=numeric_features).copy()

# Isolate features for scaling
X = df_clean[numeric_features]

# 4. Apply StandardScaler (Crucial for K-means as features have different units)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print(f"✅ Data preprocessed successfully.")
print(f"Original shape: {df.shape} | Cleaned & Scaled shape: {X_scaled.shape}")
```

✅ Data preprocessed successfully.
Original shape: (344, 7) | Cleaned & Scaled shape: (342, 4)

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# TASK 2: DETERMINING OPTIMAL CLUSTERS (ELBOW METHOD & SILHOUETTE)
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sse = []
silhouette_scores = []
k_range = range(2, 7)

# Compute metrics across a range of potential cluster sizes
for k in k_range:
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42)
    kmeans.fit(X_scaled)
    sse.append(kmeans.inertia_)
    silhouette_scores.append(silhouette_score(X_scaled, kmeans.labels_))

# Plot 1: The Elbow Method Curve (SSE vs. Cluster Count)
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.plot(k_range, sse, marker='o', color='#2B6CB0', linewidth=2)
plt.title('Elbow Method: SSE vs. Cluster Count ($k$)', fontsize=12, fontweight='bold')
plt.xlabel('Number of Clusters ($k$)')
plt.ylabel('Sum of Squared Errors (SSE / Inertia)')
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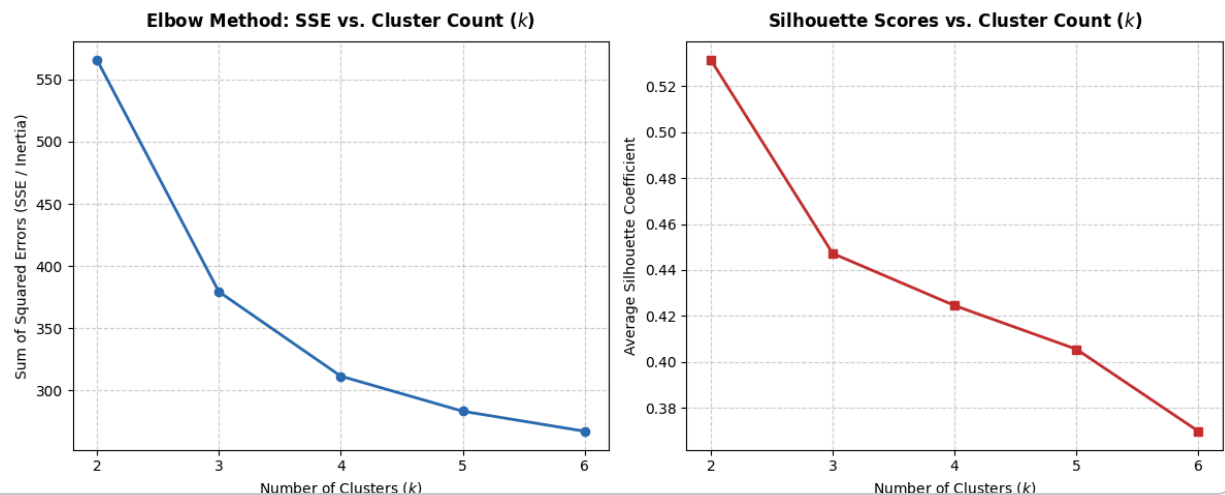
plt.xticks(k_range)
plt.grid(True, linestyle='--', alpha=0.6)

# Plot 2: Silhouette Score Curve
plt.subplot(1, 2, 2)
plt.plot(k_range, silhouette_scores, marker='s', color='#C53030', linewidth=2)
plt.title('Silhouette Scores vs. Cluster Count ($k$)', fontsize=12, fontweight='bold')
plt.xlabel('Number of Clusters ($k$)')
plt.ylabel('Average Silhouette Coefficient')
plt.xticks(k_range)
plt.grid(True, linestyle='--', alpha=0.6)

plt.tight_layout()
plt.show()

# Print metrics summary table
print("--- 📊 Cluster Validation Metrics Summary ---")
for k, s, sil in zip(k_range, sse, silhouette_scores):
    print(f"Clusters k={k} | SSE: {s:.2f} | Silhouette Score: {sil:.4f}")

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# =====
# COMBINED TASK 3 & 4: MODEL TRAINING, PLOTTING, AND EVALUATION
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# 1. Fit the final K-means model (k=3)
optimal_k = 3
final_kmeans = KMeans(n_clusters=optimal_k, init='k-means++', random_state=42)

# Save the predictions with a safe, lowercase column name 'cluster'
df_clean['cluster'] = final_kmeans.fit_predict(X_scaled)

# 2. Render the 2D Spatial Scatter plot
plt.figure(figsize=(8, 6))
sns.scatterplot(
    data=df_clean,
    x='flipper_length_mm',
    y='body_mass_g',
    hue='cluster',
    palette='Set1',
    s=70,
    alpha=0.8,
    edgecolor='black'
)

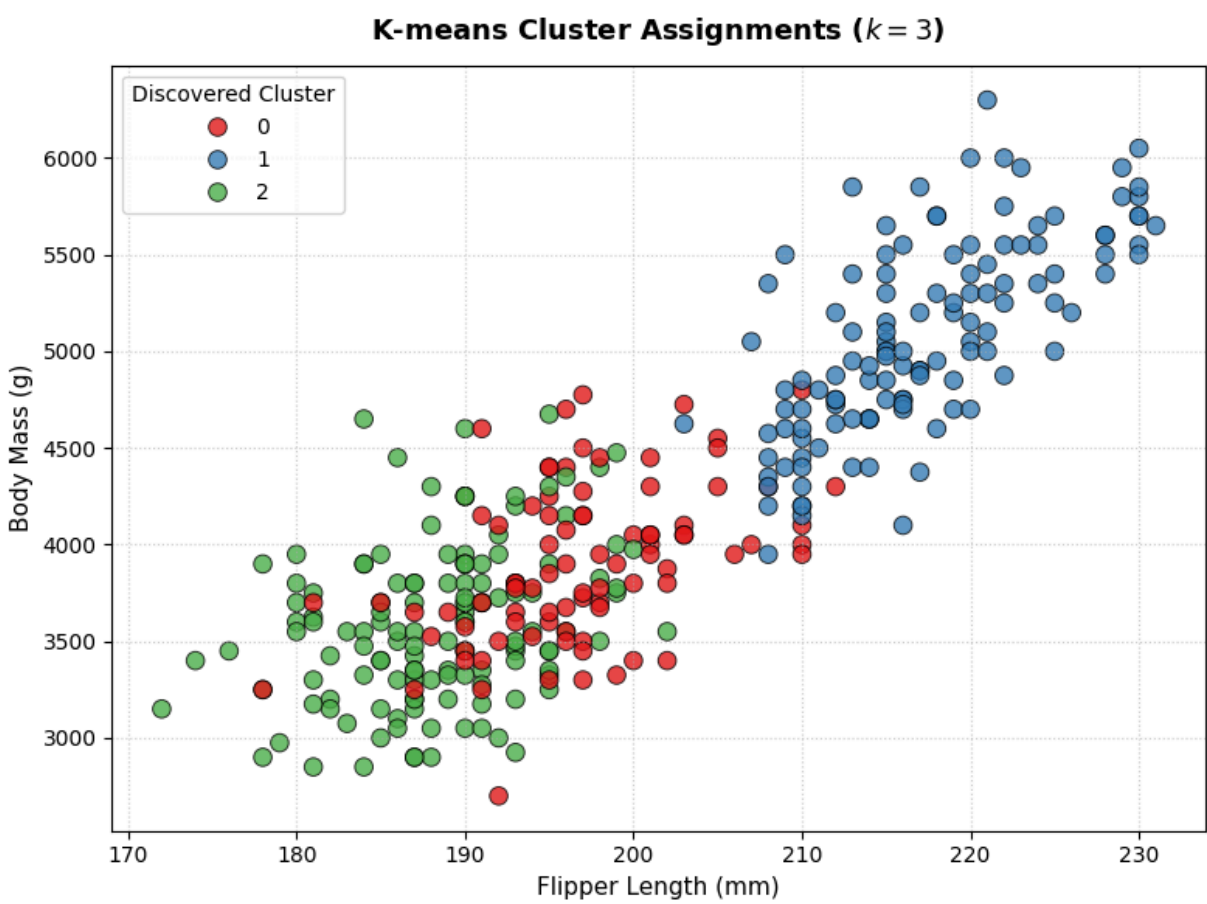
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plt.title(f'K-means Cluster Assignments ($k={optimal_k}$)', fontsize=13, fontw
plt.xlabel('Flipper Length (mm)', fontsize=11)
plt.ylabel('Body Mass (g)', fontsize=11)
plt.legend(title='Discovered Cluster', loc='upper left')
plt.grid(True, linestyle=':', alpha=0.6)
plt.tight_layout()
plt.show()

# 3. Print the biological cross-tabulations safely
print("--- 🐧 Cluster Assignment vs. Natural Biological Species ---")
species_cross = pd.crosstab(df_clean['species'], df_clean['cluster'])
print(species_cross)

print("\n--- 👤 Cluster Assignment vs. Biological Sex (Dimorphism Check) ---")
sex_cross = pd.crosstab(df_clean['sex'].fillna('Unknown'), df_clean['cluster'])
print(sex_cross)

print("\n--- 🎯 Centroid Morphological Characteristics (Feature Averages) ---")
centroids_real = df_clean.groupby('cluster')[numeric_features].mean()
print(centroids_real)
```



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--- 🐧 Cluster Assignment vs. Natural Biological Species ---
cluster    0    1    2
species
Adelie     24    0  127
Chinstrap  63    0    5
Gentoo      0  123    0
```

